

Quenton Ni

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EDUCATION

University of Minnesota, College of Science and Engineering

M.S. in Computer Science | GPA: 4.00

B.S. in Computer Science, Minor in Management Information Systems | GPA: 4.00

Minneapolis, MN

May 2027

May 2026

EXPERIENCE

IT Intern

Land O' Lakes

- Incoming IT Intern for Summer 2026

Arden Hills, MN

May 2026 – Present

Software Engineer (Capstone)

Medtronic

- Partnering with Medtronic engineers in a year-long academic industry collaboration to design, implement, and test a software solution in the medical technology domain.
- Define project scope, document system specifications, and ensure all design components comply with industry and regulatory standards.
- Contribute to the full Agile development lifecycle, including architecture design, coding, testing, and peer code reviews while leveraging DevOps tools such as Git and Azure DevOps.

Minneapolis, MN

Sep 2025 – May 2026

Research Assistant | NSF REU & UROP Grant Recipient

University of Minnesota-Twin Cities

- Designed and implemented a Quantization-Aware Training (QAT) pipeline to compress a cardiac signal classification model from FP32 to as low as INT2, achieving up to 30X model size reduction while preserving diagnostic accuracy.
- Engineered Depthwise Separable Convolutional architectures with <400 parameters for intracardiac electrogram (IEGM) classification, targeting deployment on ultra-constrained edge devices.

Minneapolis, MN

Sep 2025 – May 2026

Computer Science Teaching Assistant

University of Minnesota-Twin Cities

- Facilitated the education of 400+ students across Java, C systems programming, and x86-64 assembly by leading weekly labs, holding office hours, and testing course content.

Minneapolis, MN

Sep 2024 – May 2026

Machine Learning Teaching Assistant

UMN Data Science and AI Hub

- Supported PhD-led instruction for 90+ high school seniors, teaching core machine learning concepts including deep neural networks, transformers, and reinforcement learning.

Minneapolis, MN

May 2025 – August 2025

Projects

Multi-Dialect Vietnamese Text to Speech

- Fine-tuned and evaluated a multi-dialect Vietnamese TTS system using F5-TTS on ~71 hours of dialect-annotated speech across Northern, Central, and Southern Vietnamese.
- Built and presented an interactive Gradio demo for dialect controlled Vietnamese speech synthesis with generated audio playback and mel spectrogram visualization.

Gamblr

- Engineered a real-time gambling tracker using Vue.js and Firebase Firestore, implementing NoSQL data structures for seamless data synchronization.
- Developed a customizable drag and drop dashboard featuring interactive data visualization widgets for real-time spending analysis and habit tracking.
- Integrated Google Cloud Speech-to-Text via Firebase Function. and rendered a 3D slot machine simulation using Three.js.

Distributed Auction Marketplace System

- Built a container based distributed auction marketplace using Docker Compose, with a centralized controller, stateless service nodes, and replicated SQLite backed storage nodes.
- Implemented primary-backup replication with controller managed health checks and automatic primary failover to maintain availability during storage node failures.
- Designed autoscaling for the service node layer using rolling request rate monitoring and round robin routing.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, Typescript, HTML, CSS

Technologies: PyTorch, NumPy, Scikit-Learn, Transformers, Git/GitHub, Docker, Vue, Flask, Azure, Azure DevOps, SQL, PostgreSQL, REST APIs, Agile/Scrum

Relevant Coursework: Applied Machine Learning, Natural Language Processing, Algorithms and Data Structures, Operating Systems, Machine Architecture, Distributed Systems